



Effect of Batch Feeding Times on Greenhouse Gas and NH₃ Emissions During Meat and Bone Meal Bioconversion by Black Soldier Fly Larvae

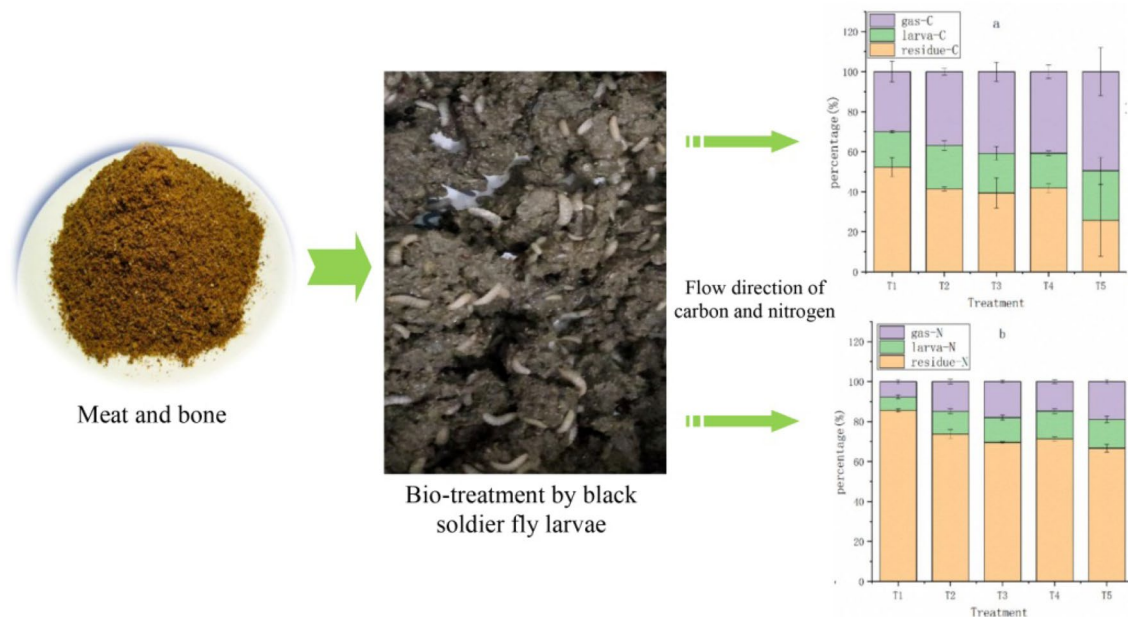
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Abstract

Meat and bone meal (MBM) is considered an organic waste with abundant nutrient elements. There is an urgent demand for new technologies to recycle MBM. MBM treatment with black soldier fly larvae (BSFL) is of greater potential to obtain more available C and N than traditional methods. This study evaluated the effect of batch feeding times on MBM bioconversion by BSFL, and investigated the emission of greenhouse gases and NH₃, the final distribution of C and N during this treatment. Our results showed that the lowest greenhouse gases and NH₃ emission was observed in 1-time batch feeding treatment. The total greenhouse gases were increased with the increasing batch feeding times, the highest emission (484.13 g CO₂-eq/kg DM) was obtained in the 5-time batch feeding treatment. The 5-time batch feeding treatment also achieved the highest substrate conversion efficiency (31.17%). Overall, using MBM as substrate for BSFL batch feeding is considered a very promising alternative.

Graphic Abstract



Keywords Batch feeding · Meat and bone meal · Black soldier fly larvae · Greenhouse gas and NH₃ · Efficiency

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Statement of Novelty

Meat and bone meal is a common biowaste. In traditional treatment methods, considerable amount of valuable C and N which are locked in the meat and bone meal are unavoidably lost from the circle. In this study, the effect of batch feeding treatment with meat and bone meal (MBM) as substrate for black soldier fly larvae (BSFL) was investigated. The results showed that changing batch feeding times can reduce greenhouse gas emission and improve carbon and nitrogen recycle from biowaste to larval biomass. Our results provide reference for using meat and bone meal as substrate for BSFL batch feeding and lay a certain technical foundation for the large scale of black soldier fly larvae bioconversion of meat and bone meal.

Introduction

There will be more than 9 billion population in the world by 2050 at the current growth rate [1]. With the increasing population, the global meat production will be doubled from the 229 million tons in 1999/2001 to 465 million tons in 2050 [2]. The intensive meat production will result in increasing slaughter wastes. Meat and bone meal (MBM) is a kind of animal waste from meat processing industry, and MBM is classified as high risk material [3]. MBM is composed of inedible livestock bodies, bones, internal organs, and other wastes, which is subjected to high temperature and high pressure processing at 133 °C for 20 min at 3 bars, and then it is crushed [4]. MBM contains a large number of elements such as C (30%), N (8%), P (5%), and Ca (10%), mainly in slowly-soluble organic forms [5]. Due to its rich nutrient elements, MBM tends to be used as crop fertilizer and solid amendment to help the growth of plants [6]. However, in traditional treatment methods, considerable amount of valuable C and N which are locked in the MBM are unavoidably lost from the circle. These lost C and N will be leached into water system or will be emitted into the air (such as GHG or NH₃, etc.). The loss of nutrients causes a decline in the quality of the end-product and environmental pollution, which prompts researchers to explore alternative treatment of MBM, such as MBM bioconversion by insect composting.

In order to enable MBM to be sufficiently utilized, it is necessary to develop economic and environment-friendly alternatives. Hence, an insect is introduced to accelerate the C and N recycle and to transfer the nutrients of MBM into insect.

Black soldier fly (*Hermetia illucens* L.) larvae (BSFL) composting is an ecologically- green technology, which

can convert various organic wastes such as livestock manure, food waste, and sludge, and the conversion residues can be utilized as bio-fertilizer. In particular, compared to other waste treatment technologies, waste-treating BSFL provides protein- and fat-rich biomass, which can be further used for biodiesel production and animal feeding, and also provides other by-products such as chitin [7] and antibacterial peptide [8]. All of these showed the potential of BSFL to recycle organic waste.

GHG and NH₃ emission is unavoidable due to many kinds of physicochemical factors in traditional composting. The percentage of N loss during the traditional composting varies from 19 to 77% with NH₃ emission as the main total N loss [9]. Besides, about 14–51% of the initial carbon content is lost in the form of greenhouse gases (GHG), particularly CH₄ and CO₂ emission [10]. Using global warming potential (GWP) to estimate GHG in composting [11], GHG contains CO₂, CH₄ and N₂O. In particular, GWP in CH₄ and N₂O was 34 and 298 folds as much as that in CO₂, respectively [12]. The emission of NH₃ is the most important pathway of N loss in composting. Literature has indicated that the BSFL composting exhibits some advantages in that the GHG and NH₃ emission in BSFL composting treatment was lower than that in traditional composting of organic waste [13]. BSFL generally exhibited a S-shaped growth curve, and BSF at different stages has different growth rates [15]. Hence, the substrate input increases with the advancing larval stage. Feeding system has been reported to have a significant influence on the larval weight [16]. However, the increasing batch feeding times lead to the labor input increase accordingly. Hence, BSFL was introduced to treat MBM by batch feeding method, the certain amount of substrate was batch fed to BSFL to explore the conversion of the substrate and the emission of GHG and NH₃. The aim of the present study was: (1) to evaluate GHG and NH₃ emission in BSFL composting treatment under different MBM batch feeding times; (2) to determine the optimal batch feeding times and BSFL conversion efficiency; and (3) to use BSFL as a bioreactor to transfer organic waste for converting the C and N from the initial substrate into larval biomass.

Materials and Methods

Materials

The experiment was performed at Huazhong Agricultural University (HZAU). The meat and bone meal was obtained from Wuhan Huanbao agriculture corporation, and the rice bran (RB) was from the experimental field of HZAU. The BSFL (*Hermetia illucens*, Wuhan strain) was provided by the State Key Laboratory of Agricultural Microbiology of HZAU, Wuhan, China. The RB was used as a kind of

loosening agent to enhance the porosity of the substrate and as a promising bulking agent so as to enhance composting efficiency. Table 1 shows the characteristics of the materials.

Experimental Design and Operation

The experiment was conducted with 3 L glass bottle reactors for 10 days. MBM and RB were mixed at 4:1 ratio on dry weight basis. The moisture content of substrate was adjusted approximately to 60%, and the total weight of substrate in each bottle was 1 kg. The substrate was provided to BSFL for different batch feeding times (5, 4, 3, 2 and 1 time) in each bottle (Table 2). Based on preliminary trials, 1500 3-day-old BSFL were put into 3 L glass bottle. The experiment was carried out in a greenhouse at 27 °C and 60–70% humidity. All treatments in this experiment were replicated three times.

Measurement of Greenhouse Gas and NH₃ Emission

The CO₂, CH₄, N₂O, and NH₃ samples were daily detected. GHG samples were collected from the headspace of each glass bottle reactor using 50-mL polypropylene syringe through the rubber tube fitted to a three-way valve, as described in previous study [14]. The collected gas samples were instantly transferred into 12-mL pre-evacuated white-cap flat-bottom glass vials (Labco Exetainer, Labco Limited, UK 839 W) for subsequent measurement of CH₄, CO₂, and N₂O. NH₃ was trapped in a sulfuric acid solution

and then measured by titration with a sodium hydroxide solution [17].

Measurement of Solid Samples

At the end of the experiment of BSFL treatment, residue and larvae were separated and weighed. All the solid samples were air dried and then crushed for the determination of Kjeldahl nitrogen (TKN) and total organic carbon (TOC). The TKN and TOC contents were analyzed using the methods of the Chinese national standard (NY525-2012) with dried samples.

Statistical Analysis

The experimental data were analyzed by the EXCEL 2010 software and plotted by the Origin 9.0. The data from different treatments were subjected to one-way analysis of variance (ANOVA) using SPSS (SPSS Statistical IBM) 20.0. $P < 0.05$ was considered as significantly different between the compared groups.

Results and Discussion

Effect of Batch Feeding Times on Gas Emission During BSFL Bioconversion

CH₄ and CO₂ Emission During BSFL Bioconversion

Daily CH₄ emission and cumulative CH₄ emission in the five treatments are presented in Fig. 1a and b, respectively. The T3 (treatment of 3-time batch feeding), T4 (treatment of 4-time batch feeding), and T5 (treatment of 5-time batch feeding), exhibited similar pattern during BSFL composting. CH₄ emission peaked on the 2nd day and fluctuated until the 6th day, and then became stable until the end of the experiment. CH₄ emission in T1 and T2 treatments first increased, peaked on the 4th day, and then dropped rapidly. CH₄ was produced from anaerobic condition [18]. The cumulative CH₄ emission amounts were 4.89, 21.34, 37.75, 110.79, and 68.31 g/kg in T1–T5 treatments, respectively. In the initial stage, the substrate contained large particles,

Table 1 Physiochemical characteristics of raw materials

	TOC(g/kg)	TKN(g/kg)	Moisture content (%)
MBM	471.2	114.9	8.5
Rice bran	484.3	4.4	10

TOC total organic carbon, TKN total Kjeldahl nitrogen

Table 2 BSFL specific feeding conditions

Treatment	No. of batch feeding times	The percentage of substrate fed on trial days									
		0	1	2	3	4	5	6	7	8	
T1	1	100%									
T2	2	50%					50%				
T3	3	33%				33%			33%		
T4	4	25%			25%		25%		25%		
T5	5	20%		20%		20%		20%		20%	

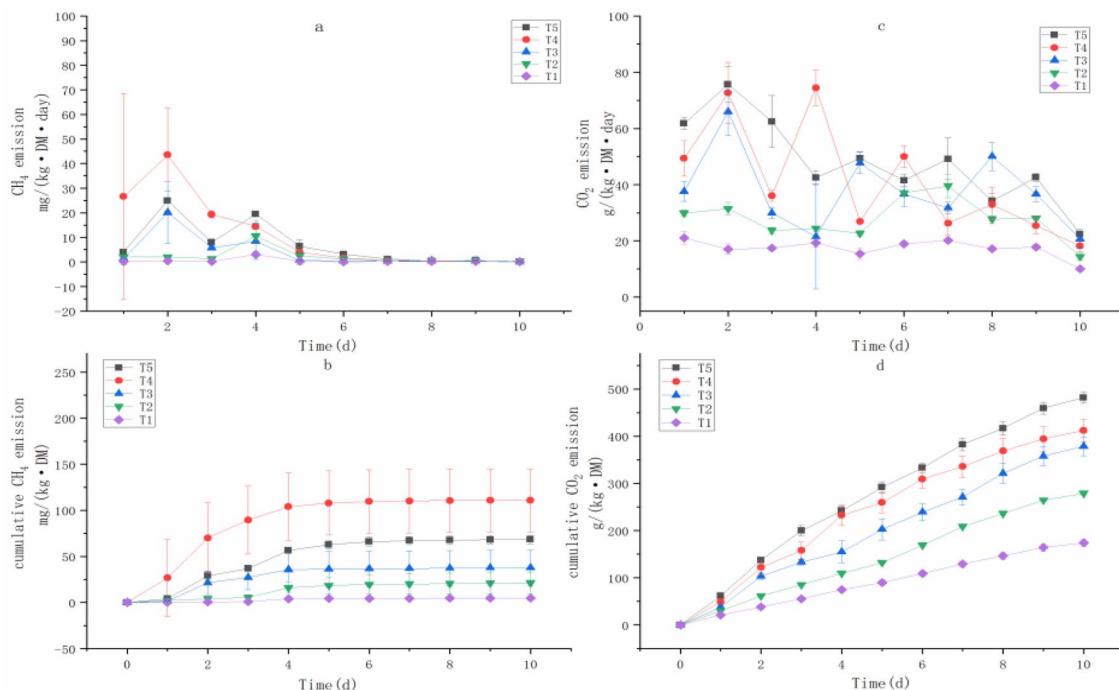


Fig. 1 Changes in daily CH₄ (a) and cumulative CH₄ (b), daily CO₂ (c), and cumulative CO₂ (d) emission during MBM bioconversion by BSFL. The data are presented as the mean of three replicates and error bars indicate standard deviation

so that the substrate was not completely covered by BSFL activity range. Thus part of uncovered substrate area will be in an anaerobic state, which was the major reason for CH₄ production. As the composting continued, BSFL promoted air circulation and broke up MBM large particles by continuously moving and degrading MBM, thereby destroying anaerobic areas, which explained the CH₄ reduction. The maximum cumulative CH₄ emission was observed in T4 treatment, while minimum cumulative CH₄ emission was found in T1 treatment. The BSFL activity mainly focused on the fresh material. Adding new material led to the formation of anoxic environment under the substrate surface, in turn resulting in an increase in CH₄ emission with the increase in feed batch times.

CO₂ is the major gas product resulting from BSFL treatment of bio-waste, which is related to the respiration of associated microorganisms and BSFL, thus the emission of CO₂ can indirectly indicate the biodegradation rate of the substrate [24][24][24]. The daily and cumulative CO₂ emission in the five treatments were presented in Fig. 1c, d. Obviously, the CO₂ emission was much higher than CH₄ and N₂O emission. In T1–T5 treatments, we observed the same daily CO₂ emission pattern that CO₂ was first increased and then decreased with addition of fresh materials.

In five different treatments, the highest peak of daily CO₂ emission occurred at different day (on the 1st, 7th, 2nd, 4th, 3rd day) with peak values of 21.08, 39.54, 69.93, 74.46,

and 62.46 mg/(kg·DM·day), respectively. The cumulative CO₂ emission amounts were 5.29, 10.55, 3.86, 10.64, and 14.93 g/(kg·DM), respectively in T1–T5. Our data indicated that CO₂ emission was positively correlated with the batch feeding times. CO₂ emission was mainly due to the respiration of organisms in MBM composting. In addition, the cumulative CO₂ emission amount was positively related with BSFL biomass in all treatments. As the batch feeding times increased, the total CO₂ emission amount and the biomass of harvested BSFL increased. One previous study reported that the CO₂ emission could reflect microbial and earthworm activities, and indicate the stabilization of substrates [19].

Since the initial weight of the larvae was the same, and the total substrate amount was the same in each treatment, the more batch feeding times led to the more competition between BSFL and microbes for the substrate, resulting in the increased CO₂ emission. Different batch feeding times could optimize nutrients in the substrate, thus contributing to improving the activity of BSFL and conversion efficiency.

NH₃ and N₂O Emission During BSFL Bioconversion Process

N₂O was reported to be produced under both aerobic and anaerobic conditions by two possible mechanisms, namely, nitrification and denitrification [20]. The daily N₂O emission in all the treatments were shown in Fig. 2a. Our data indicated that T1 reached peak on the 8th day with peak

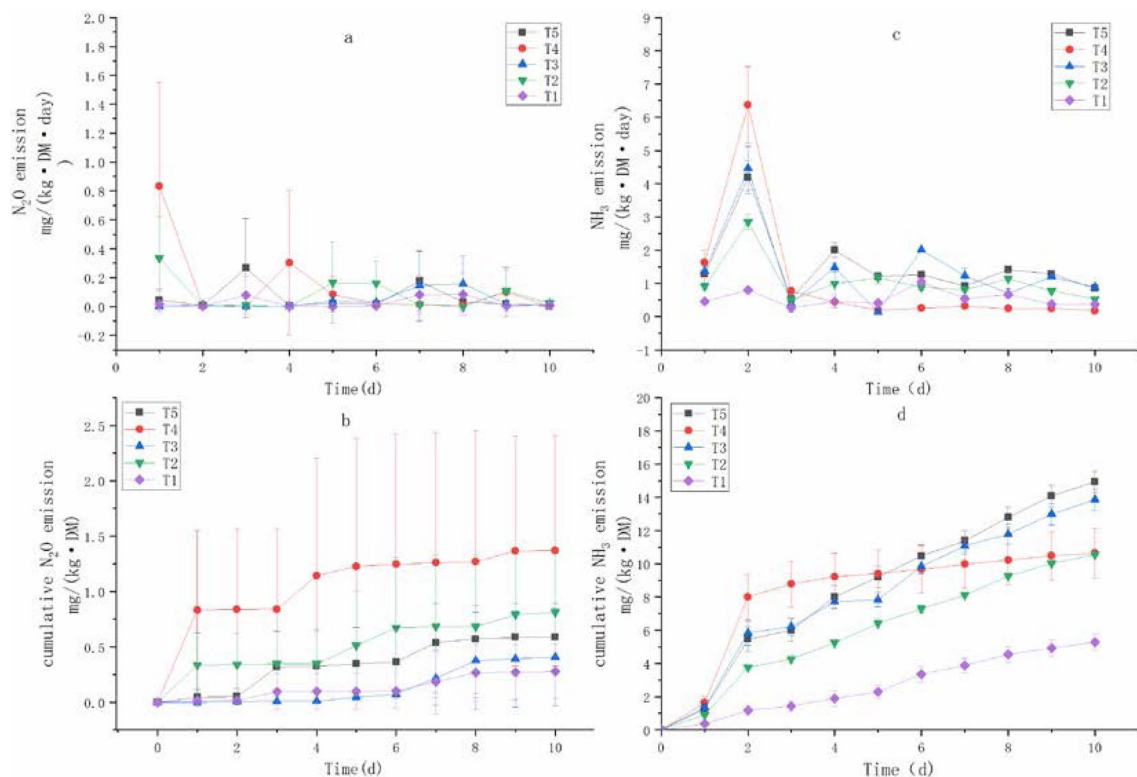


Fig. 2 Changes in N₂O (a) and cumulative N₂O (b), NH₃ (c), cumulative NH₃ (d) emissions during MBM bioconversion by BSFL. The data are expressed by the means of three replicates, and error bars indicate standard deviation

value of about 0.08 mg/(kg. DM. day), whereas T2 and T4 reached peak on the 1st day with peak value of 0.33 and 0.83 mg/(kg. DM. day). Our data also showed that T3 and T5 reached peak on the 8th and the 3th day with peak value of 0.16 and 0.27 mg/(kg. DM. day). N₂O emission amount in all treatments was very low during the BSFL bioconversion period, which led to a high standard deviation of N₂O emission. No significant difference in cumulative N₂O emission amount was observed among 5 treatments ($P > 0.05$) with the cumulative emission amounts in all 5 treatments of less than 0.27 mg. Kg⁻¹ (Fig. 2b).

Our data indicated that a high N₂O emission was observed at the beginning days (from day 1–3), and then N₂O emission decreased gradually. Such a change pattern might be attributed to the fact that the initial temperature and O₂ concentration were suitable for nitrification in the substrate during the composting [21]. However, BSFL movement increased system temperature, and ceased nitrification. NH₃ emission was usually the main nitrogen loss pathway in traditional composting. Excessive NH₃ emission could cause bad smell and adversely affect human health. NH₃ emission usually appeared in the thermophilic stage during composting process [22]. NH₃ was produced by converting organic nitrogen in the matrix into NH₄⁺, and NH₄⁺ was not converted into NO₃⁻ by nitrifying bacteria, so that NH₄⁺ was

converted to NH₃ at a certain temperature, thereby causing loss of organic nitrogen in the matrix.

NH₃ emission and cumulative NH₃ emission in the 5 treatments were exhibited in Fig. 2c, d. NH₃ emission in all treatments was rapidly increased at initial stage (on day 1–2). The maximum NH₃ emission in T2 to T5 occurred on the 2nd with the peak value of 2.84, 4.46, 6.37, and 4.19 mg/(kg.DM.day), respectively, while in T1 treatment, NH₃ emission reached the peak on the 6th day with the peak value of 1.04 mg/(kg. DM. day). In T4, NH₃ emission reached the peak on the 2nd day, and then gradually decreased until the last day (on the 10th day). Generally, high NH₃ emission was considered to be associated with high temperature and high O₂ concentration [23]. High NH₃ emission in initial stage was due to the high nitrogen content in raw substrate, but the BSFL movement made the substrate contact with air and lead to its exposure to aerobic environment. And then microbiological degradation produced a large amount of NH₄⁺ and heat, thus increasing substrate temperature and finally enhancing NH₃ volatilization. In aerobic composting, high O₂ concentration could promote NH₄⁺ transformation into NO₃⁻, and the movement of BSFL provided high O₂ concentration which was a significant factor to decrease the NH₃ emission. Another factor to reduce the NH₃ emission was moisture which can reduce the free air space in

Table 3 Characteristics of final products

	Fresh larval weight (g)	Residue weight (g)	Residue moisture content (%)
T1	66.41 ± 11.45	701.91 ± 10.84	57.78 ± 4.00
T2	99.23 ± 3.32	649.09 ± 11.32	55.63 ± 3.98
T3	122.50 ± 11.05	610.83 ± 17.25	48.01 ± 4.92
T4	125.27 ± 9.33	557.06 ± 17.84	48.49 ± 0.87
T5	147.11 ± 19.10	576.88 ± 13.70	48.68 ± 2.32

the composting system. As Table 3 shown, the final moisture contents of substrate were different in the five treatments. T1 treatment exhibited the highest moisture content ($57.78 \pm 4.00\%$). Based on it, it could be inferred that more NH_3 was absorbed in the substrate in T1, and this inference was confirmed by the data shown Fig. 2d.

In BSFL composting system, the cumulative NH_3 emission reflected the N-loss. The cumulative NH_3 emission from T1 to T5 treatments was 5.29, 10.55, 13.86, 10.64, and 14.93 mg/(kg.DM), respectively. As the batch feeding times increased, the cumulative NH_3 emission increased.

Carbon and Nitrogen Balance During BSFL Composting

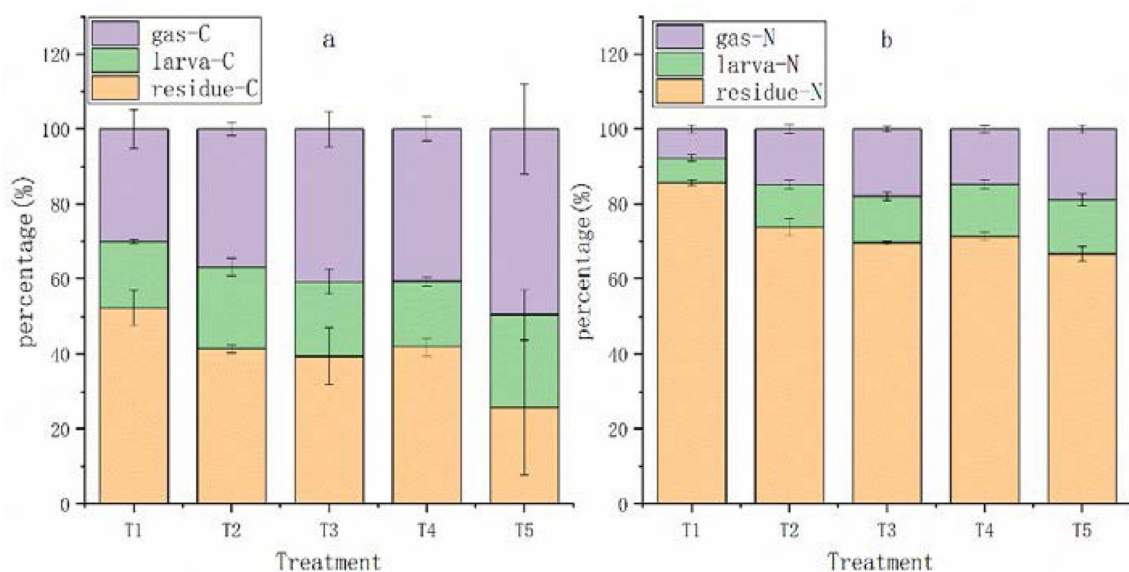
Evaluation of Carbon Transformation

To evaluate C balance, total organic carbon (TOC) in initial substrate and final substrate was measured. The cumulative C loss through CO_2 emission accounted for 13.30–36.76% of

the initial TOC due to BSFL and microbial metabolism in a natural process. CO_2 emission causing C loss was increased with the increased batch feeding times, which was attributed to the fact that the aeration rate and microbial activity was increased with the movement of BSFL in the substrate. In addition, BSFL activity decreased CH_4 emission, and almost eliminated the effect of CH_4 (0.00–0.02%). The BSFL obtained its nutrients by converting the substrate. In substrate conversion, larva-C accounted for 7.82–18.55% of the total carbon. Furthermore, the BSFL in the T5 treatment exhibited the highest C content, indicating that more C was transferred to the BSFL from substrate, which meant a higher C conversion efficiency. The residue-C accounted for 10.55 to 32.45% of total C in 5 treatment, and the residue C in T1 treatment displayed the lowest C content (10.55%), and the other C losses in this treatment accounted for 68.33%, indicating that most C was lost in other forms during treatment process and less C could be transformed into the gases or the BSFL biomass.

Evaluation of Nitrogen Transformation

To evaluate N balance, the total kjeldahl nitrogen (TKN) in the initial and final substrate was measured, and the flow of N was observed. Figure 3b showed the N distribution, including N gas, nitrogen content in the residue, nitrogen content in BSFL and other nitrogen. As a major pathway of gaseous N loss, NH_3 emission accounted for 4.08–11.49% of the initial TKN in 5 treatments. Similar to the CH_4 emission, N_2O emission accounted for only < 0.01% of the initial TKN in all 5 treatments. Due to the BSFL activity, N_2O emission was significantly reduced, which even eliminated

**Fig. 3** Carbon (a) and nitrogen balance (b)

its negative effects. As batch feeding times increased, the N loss in the form of NH_3 emission also increased. After the substrate was converted by BSFL, the nitrogen content in the larvae accounted for 3.57–8.71% of the total N content in all the 5 treatments, and the highest N content of the BSFL (8.71%) was observed in the T5 treatment, indicating that the more N was transferred from substrate to BSFL with the increased batch feeding times. The other N loss accounted for 39.19–46.98%, and the N loss in T1 was the highest, implying that the substrate N could not be converted well in T1 treatment since most N was lost in other forms, and could not be transferred to BSFL.

Characteristics of Final Products

Table 3 displayed the fresh larval biomass and final residue weight. Significant differences ($P < 0.05$) in fresh larval mass were found among the five treatments. The highest larval weight was observed in T5 treatment (147.11 ± 19.10 g), followed by T4 (125.27 ± 9.33 g), T3 (122.50 ± 11.05 g), T2 (99.23 ± 3.32 g), and T1 (66.41 ± 11.45 g). However, in all the 5 treatments, residue weight decreased with the increase in batch feeding times, implying that increasing the batch feeding times could accelerate the decomposition of substrates. Table 3 also showed that the residue moisture content. Little difference in residue moisture was observed among T3, T4, and T5, all of which was about 48%, while in T1 and T2 treatments, residue moistures were 57.78% and 55.63%, respectively, indicating that there was a negative correlation between the batch feeding times and moisture content.

Table 4 showed the C and N conversion efficiency in the harvested larvae and residue. In all the 5 treatments, the C + N in the larvae increased from 6.79 g/kg to 27.28 g/kg, with the increased batch feeding times, while the C + N in the residue ranged from 30.58 g/kg to 72.72 g/kg and the highest C + N (72.72 g/kg) was observed in the T4 treatment residue. The C + N conversion efficiency was increased with the increased batch feeding times in four treatments with exception of the fluctuation in T3. The maximum C + N conversion efficiency (31.17%) was observed in T5 (Table 4). To calculate GHG, the GHG was converted to CO_2 equivalent. When the global warming potential (GWP) of CO_2 was set

as 1, GWP of CH_4 and N_2O was calculated to be 34 and 298, respectively [12]. As shown in Table 4, the total GHG emission increased with the increased batch feeding times.

Practical Application and Future Research Perspective

With the increasing global consumption of livestock and poultry meat products, MBM, as a by-product of livestock and poultry products, also increases in quantity. Due to its rich nutritional content, MBM is often used as a soil amendment. Considering this, the present study made the first attempt to evaluate the MBM as a substrate for batch feeding BSFL. BSFL composting is an emerging organic waste treatment method that can convert MBM into biofertilizer and insect proteins. The times of batch feeding explored in this experiment played a role in increasing the larval biomass, or reducing the GHG production during BSFL composting process. This study will provide data for the future large-scale BSFL composting and protein production.

Conclusion

This study used MBM as a raw material for BSFL composting for the first time, BSFL acted as a green reactor to convert organic waste. Our results indicated that batch feeding times had a great impact on GHG and NH_3 emissions. The accumulative emission amount of CH_4 , N_2O , CO_2 , and NH_3 was the lowest in the one-time batch feeding treatment, which was 4.89 mg/kg DM, 0.27 mg/kg DM, 174.22 g/kg DM, and 5.29 mg/kg DM, respectively. With the increase in the batch feeding times, the highest GHG cumulative amount was achieved in T5 treatment. The total cumulative CO_2 emission amount was found to be the highest in the T5, namely, five-time batch feeding treatment, which was consistent with the largest total weight of fresh BSFL. Our data indicated that one-time batch feeding BSFL treatment achieved the lowest GHG, and that five-time batch feeding achieved the highest BSFL biomass. Hence, MBM treatment by BSFL can reduce GHG emission, or obtain high biomass to provide bioavailable protein and lipids for animal feeding. This study provides a novel information on the use of

Table 4 Direct C + N in larva, residue and global warming potential

	Larva C + N(g/kg)	Residue C + N(g/kg)	C + N efficiency (%)	GHG emission equivalent(g CO_2 -eq/kg DM)			
				CH_4	N_2O	CO_2	Total GHG
T1	6.79	30.58	18.17	0.1664	0.0830	174.22	174.47
T2	17.68	60.59	22.59	0.7258	0.2424	278.92	279.88
T3	20.88	67.80	23.54	1.2836	0.1209	378.50	379.91
T4	21.32	72.72	22.67	3.7672	0.4081	412.35	416.53
T5	27.28	60.24	31.17	2.3226	0.1752	481.63	484.13

meat and bone meal as substrate for BSFL composting, and lays a certain technical foundation for the large-scale BSFL conversion of meat and bone meal.

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